

Mapping Vote Swing Between Elections, 2020–2024

Every county's move from 2020 to 2024, drawn as three thousand arrows

Democracy's Data

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The country moved right in 2024. You have heard that sentence, and a red-and-blue map of who won cannot check it. A county that went from 70% Republican to 60% Republican is still red, and it just moved ten points toward the Democrats. Color shows the level; the sentence is a claim about change. So turn it into a question a citizen can answer from the record: **which way did the country move between 2020 and 2024, and where – everywhere at once, or only in some places?**

The form that answers it is a wind map, which draws **change** instead of **level**. Every place gets an arrow: which way it moved between the two elections, and how far. Together the arrows look like a weather chart, and they read like one – you can see whether a front crossed the whole country or only part of it. That is the argument for the form. One mark carries direction and size at once, which is exactly the pair of things the question asks and exactly the pair a colored map cannot show.

01 The test

The question needs the same places observed twice, and the county is the right unit mostly by elimination. States are too coarse to show where movement happened, precincts are not published nationally at all, and counties are the only level in between with national coverage in both years.

No federal source publishes county returns. The Federal Election Commission publishes official results by state and by congressional district and not by county, and no agency assembles what the states publish. So every national county map anyone has drawn rests on somebody's private compilation. The one here is Tony McGovern's repository of county presidential returns for 2020 and 2024 (https://github.com/tonmcg/US_County_Level_Election_Results_08-24), whose own README says the figures were read off news organizations' results pages and **"are not authoritative."** A file that admits what it is makes an honest foundation, and the limits section below takes the admission seriously.

What a certified return is, who writes it and what it can bear is the subject of this cluster's returns chapter. What it costs to assemble the fifty-one states' own files instead is the county-returns brief's.

Two more sources anchor the picture. Each arrow starts at the Census Bureau's population-weighted center of population for its county – where the county's people actually are, not the middle of its shape. For a western county the difference can be a hundred kilometers of empty desert. There are 3,221 of these centers, built from the whole census rather than a sample. And Georgia is drawn a second time from the Georgia

Secretary of State's own certified returns for both years. One map from a compilation nobody certifies, one from a state's legal record, the same encoding on both: that comparison is part of the test.

02 The data

The build joins the two years of returns by county code, attaches each county's center point, and writes `wind_us.csv`, one row per county drawn. The columns that matter carry the analysis: `margin_20` and `margin_24` are the two-party margins, Republican votes minus Democratic votes divided by the sum of the two. `swing` is the difference between them, in points, positive toward the Republican. `net_votes` is the swing multiplied by the county's two-party vote — the net number of ballots that changed hands. `pop` is the county's census population, and `x`, `y`, `dx`, `dy` are the arrow's position and shape, computed once by the build so every figure draws the identical geometry.

The identifier decides everything, so `county_fips` is read and stored as text. Read 01001 as a number and it becomes 1001, which matches nothing, and every state whose code begins with zero silently drops out of the join. A join that matches nothing does not complain; it just returns fewer rows.

Cleaning here meant refusing to let the join decide anything on its own. Three places are not the same unit in the two files, and each got a decision. **Alaska** reports presidential votes by State House District, under codes that differ between the files and boundaries redrawn after the 2020 census, so it is dropped. **Connecticut** appears as eight counties in the 2020 file and, in the 2024 file, as the nine planning regions the Census Bureau adopted as county-equivalents in 2022. The two sets of units cannot be compared, so its 1,824,280 votes are dropped. **The District of Columbia** is one row in 2020 and eight wards in 2024, under invented codes, so the wards are summed back into one city.

Hawaii lies outside the map's frame, and 3,100 counties survive to be drawn. Nothing was imputed (no missing value was filled in by guessing it from something else), and no vote total was adjusted.

`wind_ga.csv` holds the same columns for Georgia alone, built from the state's own returns. Georgia has had exactly 159 counties since 1945, so no unit changed and every join key can be checked by eye.

Before you look, commit to a share. The map below draws 3,100 county arrows for 2020 to 2024. **How many lean toward the Republican — a bare majority, about three in four, or nearly all?** And a second commitment: will the country as a whole move further than its typical county, or less far? Write both answers down.

03 What it says about democracy

First, the code the arrows are written in. It is fixed once, and every map below uses it.

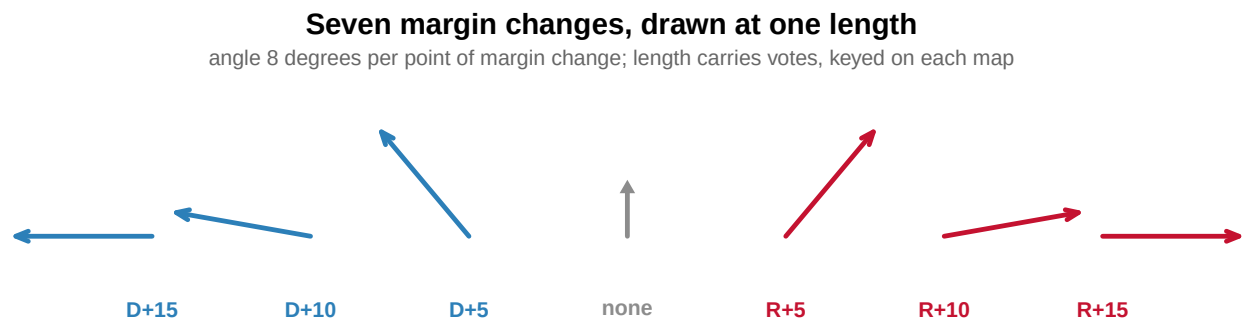


Figure 1. One arrow, seven swings – the key to every map in this brief. An arrow fits this question because one mark carries both of the question's parts. Direction is the rate: the arrow rotates 8 degrees per point of margin change, clockwise toward the Republican in red, counter-clockwise toward the Democrat in blue, and flat at 11 points. Length is the quantity: the net votes that changed hands, reaching full extension at 25,000 votes on the national map. A 15% floor keeps a small county visible, and the ceiling stops one giant county setting the scale for three thousand.

Ten points in a county of nine hundred and ten points in a county of nine hundred thousand are not the same event. That is why angle and length carry different measurements rather than saying one thing twice. Both constants are printed in each map's legend. Here is the country.

How every county moved, 2020 to 2024

two-party margin; leaning right = Republican gain, left = Democratic gain

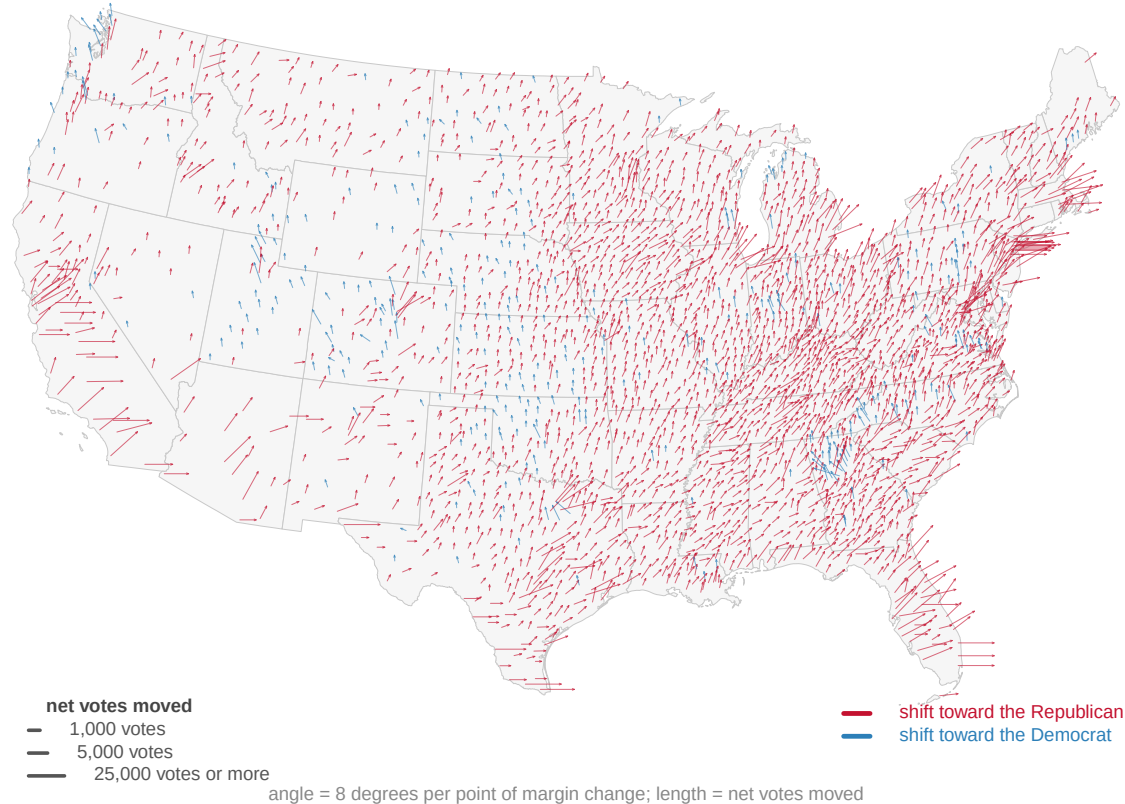


Figure 2. How every county moved between 2020 and 2024. One arrow per county, starting at the county's population-weighted center. Arrows lean right where the Republican two-party margin grew and left where it shrank, at 8 degrees per point; length is the net votes moved. 88.5% of the 3,100 arrows lean right: this is what a national swing looks like when it is close to national.

The first answer is unambiguous. **88.5% of counties moved toward the Republican candidate**, the median county by R+3.1, and the whole field leans one way. The swing was broad. It reached counties that vote Democratic and counties that vote Republican, dense places and empty ones, both coasts and the middle.

Quantity	Value
Counties drawn	3,100
Median county shift	R+3.1
Counties shifting right (%)	88.5
Largest shift right	R+28.1
Largest shift left	D+9.2
Arrows at the length cap	60

Broad, but not uniform, and the map shows both at once. 356 counties, 11.5% of the map, moved toward the Democrat anyway, against the tide. The spread among the rest is wide. The largest county move toward the Republican was R+28.1 and the largest the other

way D+9.2, while 60 arrows sit at the length cap, understating their counties' movement. A single national number contains none of that. The arrows are the evidence that "the country moved" is an average over places that moved very differently, and the length channel says where the movement carried real ballots rather than empty land.

Which raises the second question you committed to. Every arrow on that map is the same kind of mark, and the counties are not the same size. The largest cast 3,606,971 two-party votes and the smallest 96, a ratio of about 37,573 to one.

Question	Answer
What did the typical county do?	mean county shift R+3.4
What did the typical county do? (median)	median county shift R+3.1
What did the votes do?	national two-party margin moved R+6.0

The typical county moved R+3.1. The country moved R+6.0 – roughly twice as far. Both numbers are correct; they answer different questions. The gap between them means the biggest counties swung harder than the small ones, and it is why a wind map has to be read with care.

One arrow per county gives rural America several times the visual weight it has in the electorate. Newspapers make that choice constantly, usually without saying so. This brief makes it too, and the difference is that it is being said, with the vote-counted number printed beside the county-counted one.

The national map is drawn from a compilation nobody certifies. Here is the same encoding on data that is somebody's legal record.

Georgia, county by county, 2020 to 2024

Secretary of State returns, both years; 159 counties, unchanged since 1945

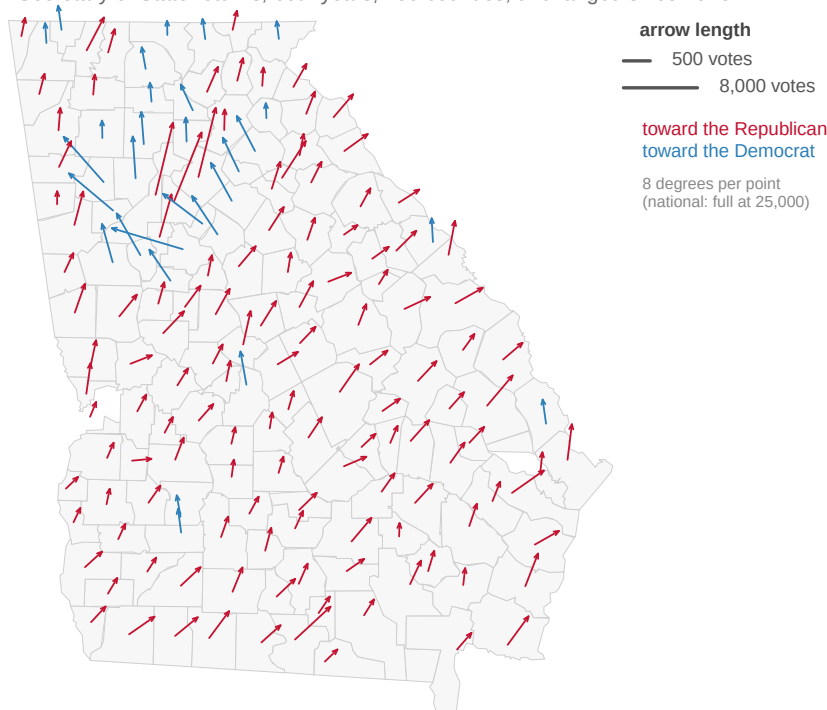


Figure 3. Georgia between 2020 and 2024, from the state's own returns. 159 counties, Secretary of State returns on both ends, on units unchanged since 1945, so no arrow here can be an artifact of a broken join. With 159 arrows on a state-sized frame there is room to read each angle: rotation is the same 8 degrees per point, recoverable with a protractor, and length reaches full extension at 8,000 net votes.

Quantity	Value
Counties drawn	159
Georgia margin, 2020	D+0.3
Georgia margin, 2024	R+2.2
Statewide shift	R+2.5
Median county shift	R+3.0
Counties shifting right (%)	81.8
Biggest shift right	Webster R+10.6
Biggest shift left	Henry D+9.2

Georgia tells the national story in miniature, from certified numbers. 81.8% of its counties leaned right, the middle one by R+3.0, and the statewide margin moved R+2.5, from D+0.3 to R+2.2. The exceptions are on the map too: Henry County moved D+9.2, the other way, while Webster County moved R+10.6 right. A press compilation and a state's certified record, drawn with one rule, point the same direction – which is the kind of agreement a claim about the country should be asked to show.

04 What this brief cannot tell you

It cannot tell you who changed their mind. A margin swing is a difference between two totals, and it ignores turnout entirely. A county can swing ten points because voters switched, because one side's voters stayed home, or because new voters arrived, and the arrow is identical in all three cases. A county whose margin held while a third of its voters stopped voting draws a stubby arrow that says nothing changed.

It cannot certify a single count. The national file is a press compilation that calls itself not authoritative, and there is no federal county file to check it against. The fifty-one state canvasses can be read one at a time, and the county-returns brief does exactly that, but no arrow here carries a certification.

It cannot draw the places its join lost. Alaska, Connecticut and Hawaii are simply absent, for the reasons given in The data, and an absence on a map does not look like a warning. It looks like nothing happened there.

It cannot express doubt. An arrow computed from 96 votes is drawn exactly as crisply as one computed from 3,606,971, and there is no obvious way to put an error bar on a direction. The direction itself is a metaphor: nothing traveled north-east, and the form borrows its persuasiveness from weather maps, where the arrows mean a thing really is moving.

And its geography is slightly stale. The center points are 2020 vintage, so the map puts 2024's voters where 2020's people lived.

05 What you should have learned

- Between 2020 and 2024, 88.5% of 3,100 counties moved toward the Republican candidate: the swing was broad, and the 356 counties that moved the other way are the exceptions, not the rule.
- The median county moved R+3.1 while the country moved R+6.0 — counting counties and counting voters are different operations, and only one of them elects anybody.
- A wind-map arrow encodes two measurements at once: 8 degrees of rotation per point of margin change, and length for the 25,000-vote scale of ballots that changed hands.
- Georgia's certified returns, on 159 counties unchanged since 1945, show the same direction as the national compilation: a statewide move of R+2.5, with 81.8% of counties leaning right.
- A join across two elections is only as good as its units: Alaska, Connecticut and the District of Columbia were not the same units in both files, and 1,824,280 Connecticut votes are missing from the map because of it.

06 Extensions

None of these is answered above, and every one can be answered by sorting or filtering the tables the Sources section hands over.

- `wind_us.csv` has `swing` for every county drawn. Sort it ascending and read the `state_name` column of the counties that moved toward the Democrat. Do they cluster in a few states, or scatter?
- `wind_us.csv` has both `swing` and `net_votes`. Sort by each in turn. Are the counties that moved the most points the same counties that moved the most ballots, and what does the difference between the two lists tell you?
- `wind_us.csv` carries `margin_20` and `margin_24`. Filter for counties where the sign changed — counties that changed winner. How many are there, and how big was the typical one's swing compared with the map-wide median?
- `wind_ga.csv` has `swing` and `net_votes` for every Georgia county. Which county moved the most ballots, and is it also the one that moved the most points?

Two stretch ideas point past the spreadsheet. The MIT Election Data and Science Lab publishes county returns built from official state records back to 2000 (<https://github.com/MEDSL/county-returns>). Download 2016 and ask whether the previous pair of elections leans the same way this one does. And pick one state, download its Secretary of State's certified county totals, and check them against the compilation's counts in `wind_us.csv`. Any disagreement you find is the price of the only national file there is.

07 Sources

Data

- County presidential returns, 2020 and 2024 — Tony McGovern, *US County Level Election Results 08-24*. https://github.com/tonmcg/US_County_Level_Election_Results

_08-24 **Not authoritative, by its own account** — the README states the figures were copied by a program from news organizations' results pages. Every arrow on the national map is drawn from it, on purpose: no federal file exists to replace it with.

- U.S. Census Bureau, *Centers of Population by County: 2020*, downloaded 2026-08-10. https://www2.census.gov/geo/docs/reference/cenpop2020/county/CenPop2020_Mean_CO.txt
- Georgia Secretary of State, *Election Results*. November 3 2020 general election archive and the November 2024 results export. <https://sos.ga.gov/page/historical-elections-results> and <https://results.sos.ga.gov/>
- Federal Election Commission, *Election Results and Voting Information* — official federal returns, published by state and congressional district and **not by county**, which is why the compilation above has to exist. <https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/>
- **Build.** `data/build-data.R` downloads the sources and writes the tables below. The encoding constants (8 degrees per point, full length at 25,000 net votes nationally and 8,000 in Georgia, a 15% length floor) are declared once in `data/encoding.R` and consumed by every figure, so none is derived from the data it draws.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`wind_us.csv`, `wind_ga.csv`, `county_centroids.csv`, `us_outline.csv`, `ga_outline.csv`

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE FIRST SOURCE. County-level presidential returns for 2020 and 2024 from Tony McGovern's compilation, served raw from GitHub:

https://raw.githubusercontent.com/tonmcg/US_County_Level_Election_Results_08-24/master/2020_US_County_Level_Presidential_Results.csv

and the matching 2024 file beside it. Each row is one county, keyed by FIPS code, with votes by candidate. No account or key is needed. Its own README says the numbers were scraped from news organizations and are not authoritative — there is no federal publisher of county presidential returns, and that is part of the story.

THE SECOND SOURCE. The Census Bureau's population-weighted county centroids, one plain text file:

https://www2.census.gov/geo/docs/reference/cenpop2020/county/CenPop2020_Mean_CO.txt

Each row is a county with the longitude and latitude of its people, not of its empty land. The header carries an invisible byte-order mark; read it as UTF-8-with-BOM or the first column name arrives

mangled.

THE THIRD SOURCE. For Georgia alone, the state's own certified returns, exported as JSON by the Secretary of State:

<https://results.sos.ga.gov/cdn/results/Georgia/export-2024NovGen.json>

WHAT TO BUILD.

1. A national table of county vote swings, 2020 to 2024, joined to the centroids by FIPS. Do not trust the join: Alaska reports State House districts under invented FIPS codes, three of which collide with real borough codes; Connecticut swapped eight counties for nine planning regions between the files; and the code for the District of Columbia means the whole city in one file and a single ward in the other. A naive join succeeds everywhere and is wrong in all three places.

2. The same table for Georgia's 159 counties from the state's own export – the clean version, where every join key can be checked by eye.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 3,221 county centroids in the census file
- 3,152 rows in the 2020 returns and 3,160 in 2024, of which only 3,104 describe the same place in both years
- 159 Georgia counties, joining both years with zero misses

The GitHub compilation takes corrections as counties certify and amend; small drifts in its row counts are revisions, not errors.